

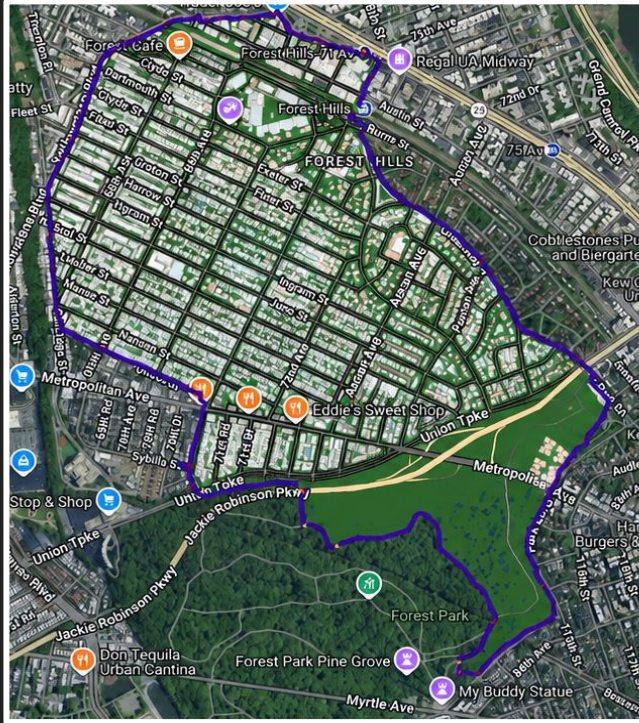
Urban Climate Analysis

Forest Hills/Kew Garden

Nabidul Islam

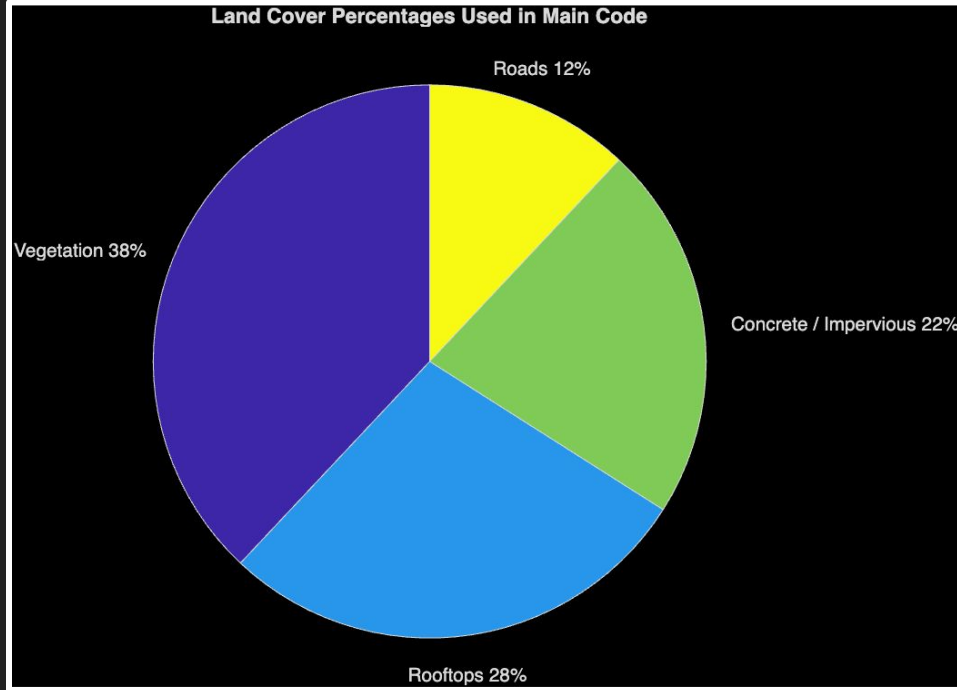


Land Cover Analysis



White represents roofs/buildings, green represents vegetation, black is asphalt, and gray is concrete.

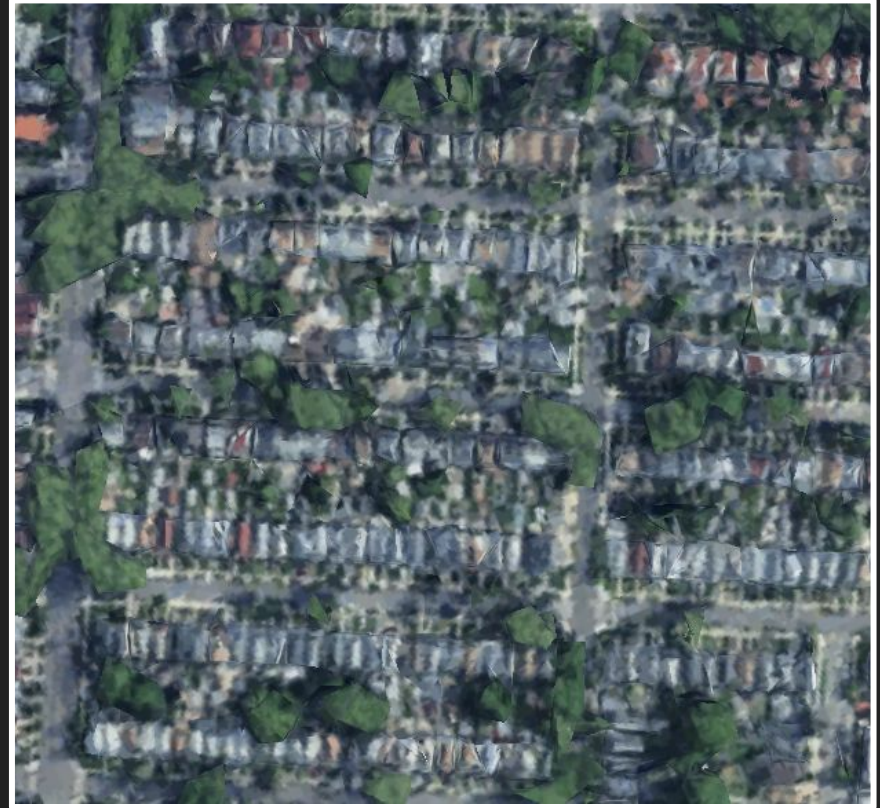
Land Cover Analysis



The land cover percentage was determined to be approximately 12% roads, 22% concrete, 28% rooftops/buildings, and 38% vegetation.

The average building height was determined to be approximately 10 meters.

Land Cover Analysis



Land Cover Analysis

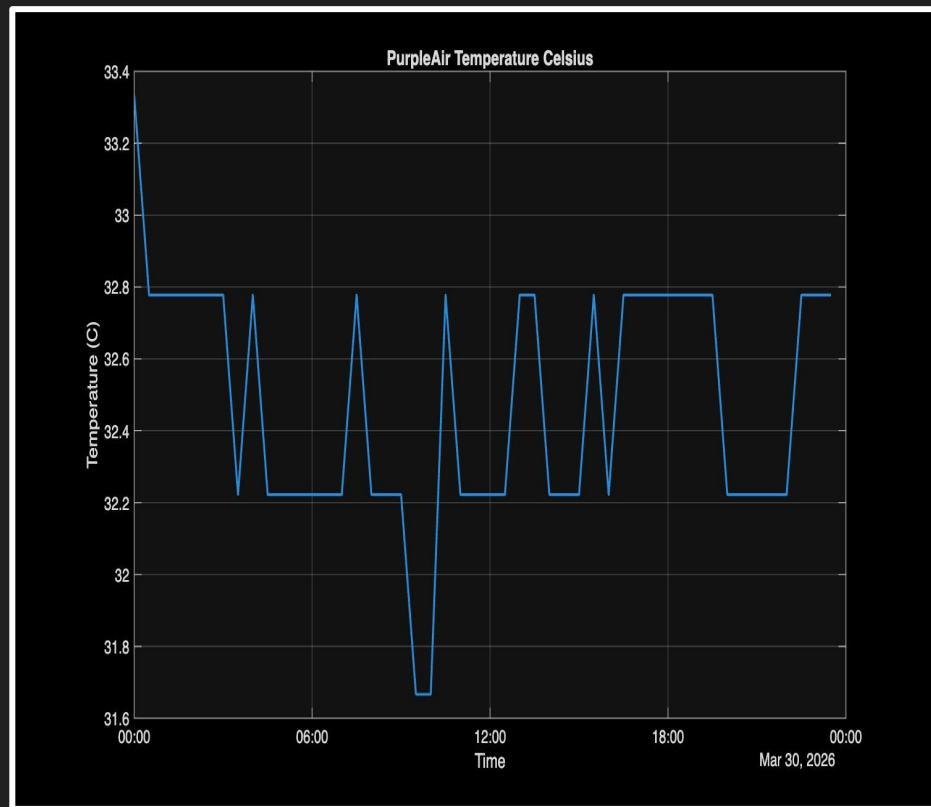
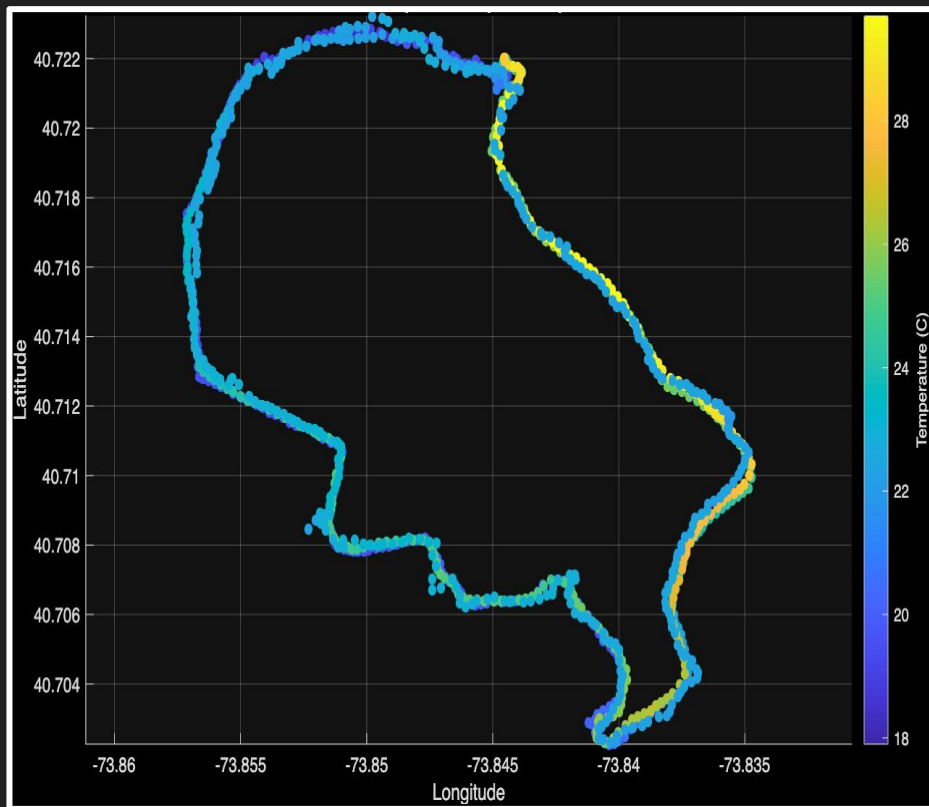
The area walked was really diverse in terms of the types of buildings and layout. To accurately determine the Land Climate Zones, parts of it need to be split up.

The first figure can best be said to be LCZ 6 - Open Low Rise. The buildings there were largely one family and were spread further out.

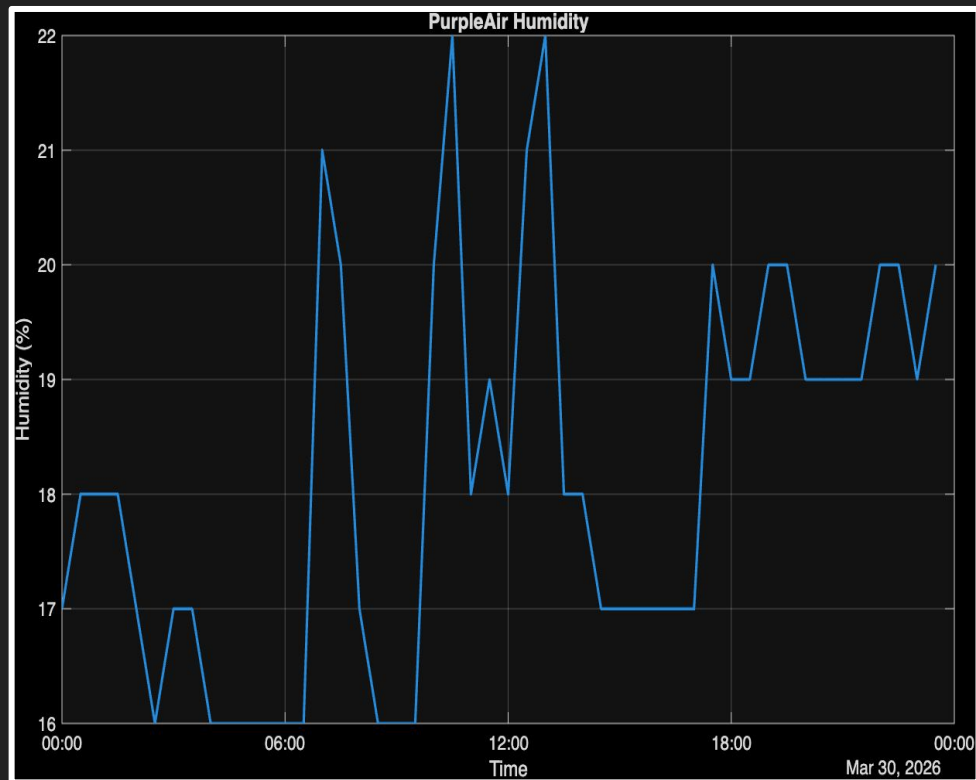
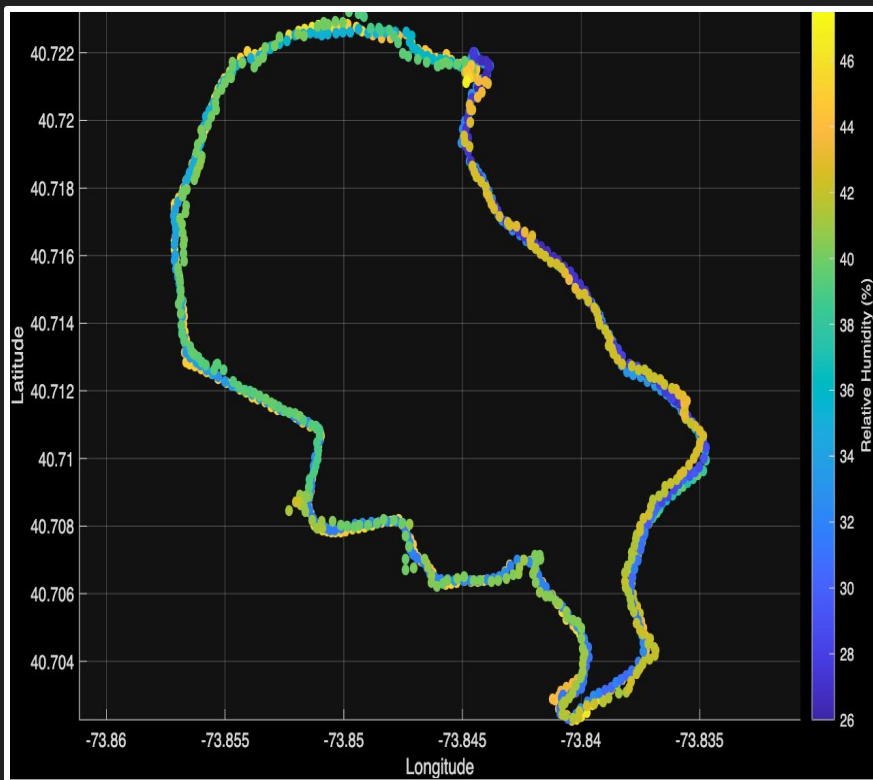
In the second figure, it can be said to be LCZ 3 - Compact Low Rise. Most of the buildings were one-to-two stories and were very tightly packed together.

Furthermore, there was also a large park that dominated a good portion of the route that can be said to be LCZ A.

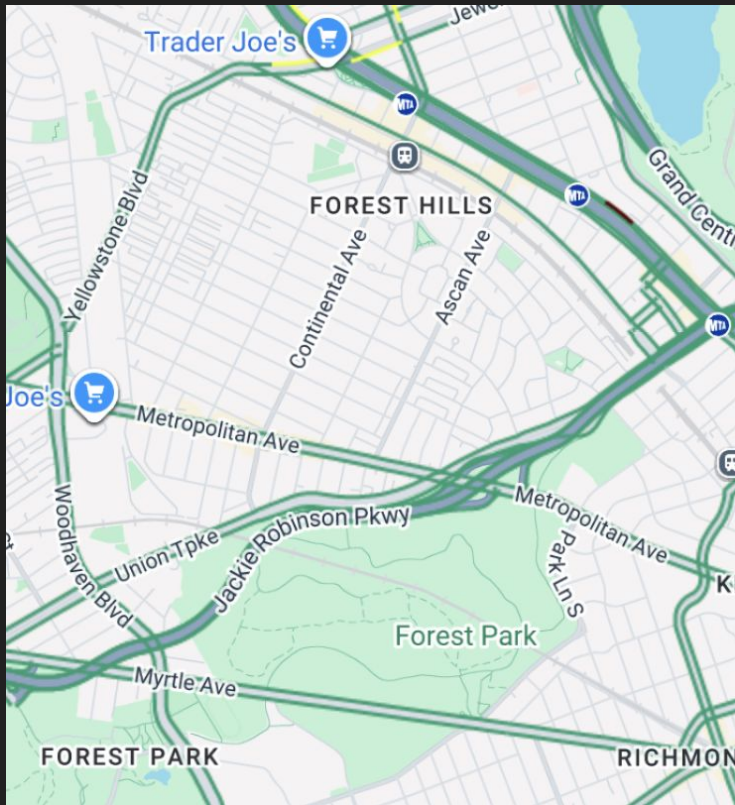
Temperature Spatial Graph - PM



Humidity Spatial Graph - PM



Traffic Data

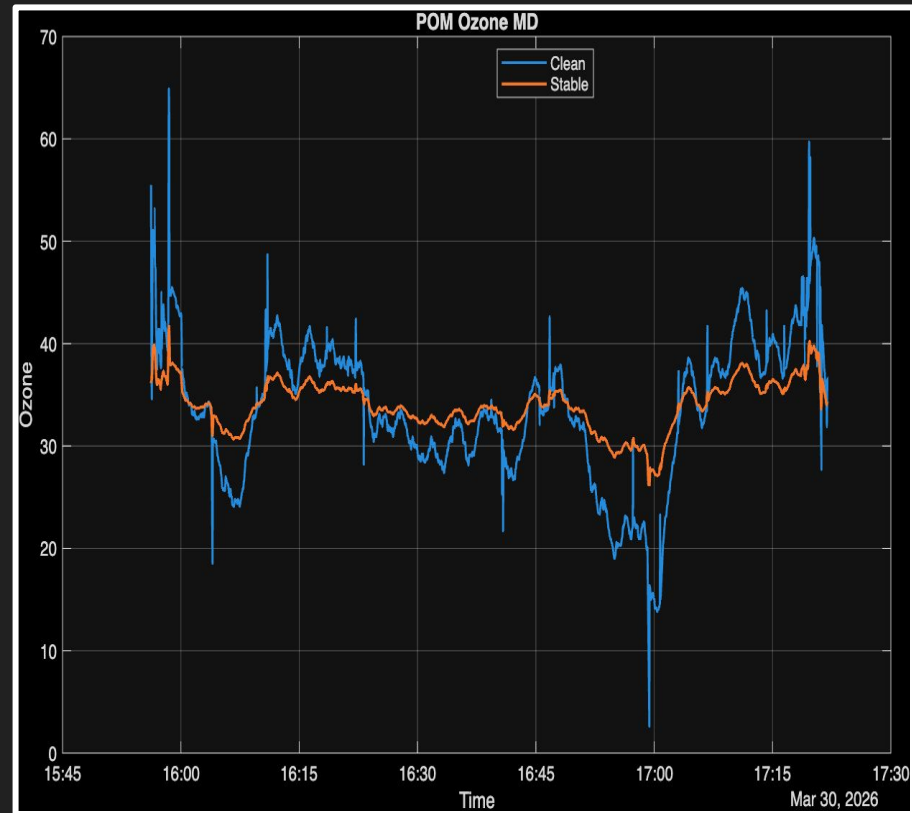


-333,000 cars on the Grand Central Parkway everyday.

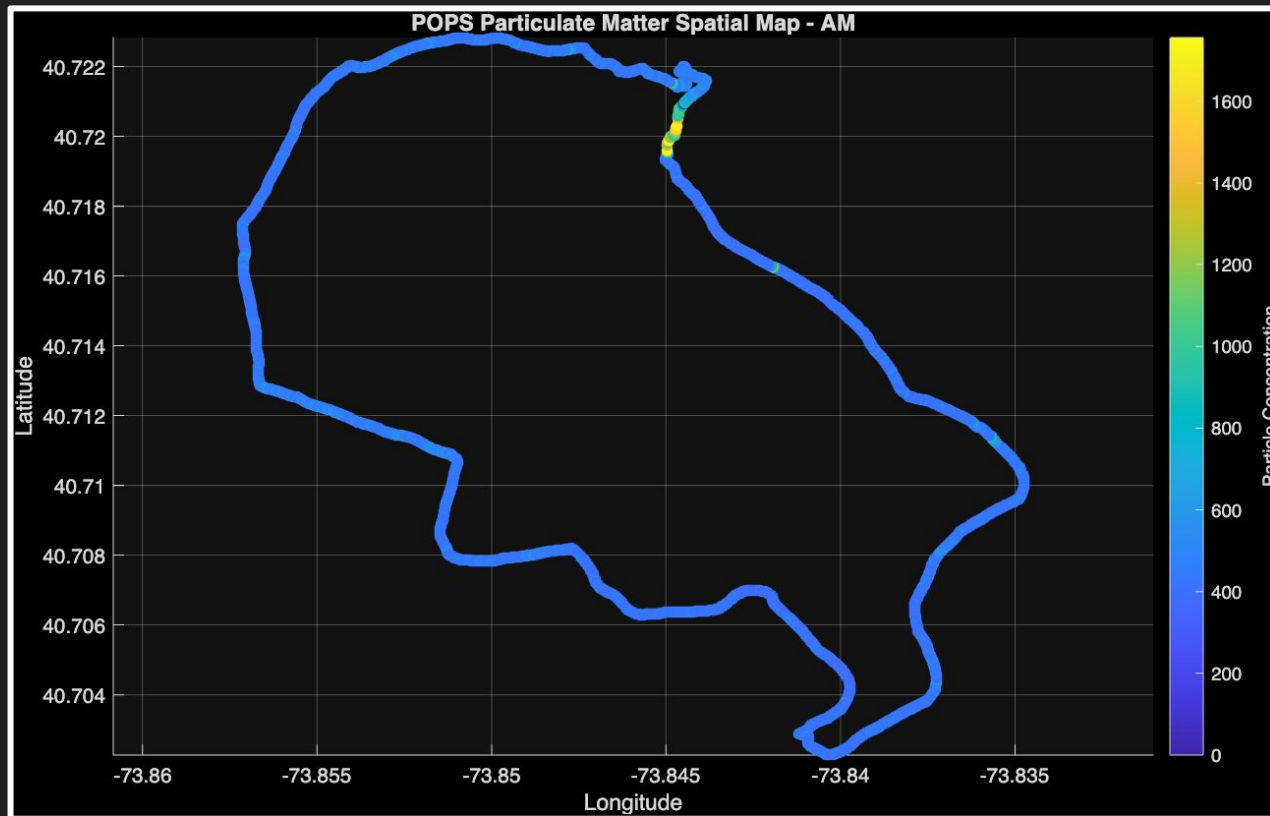
-85,000 cars on Queens Blvd

-70,000 cars on the Jackie Robinson Parkway

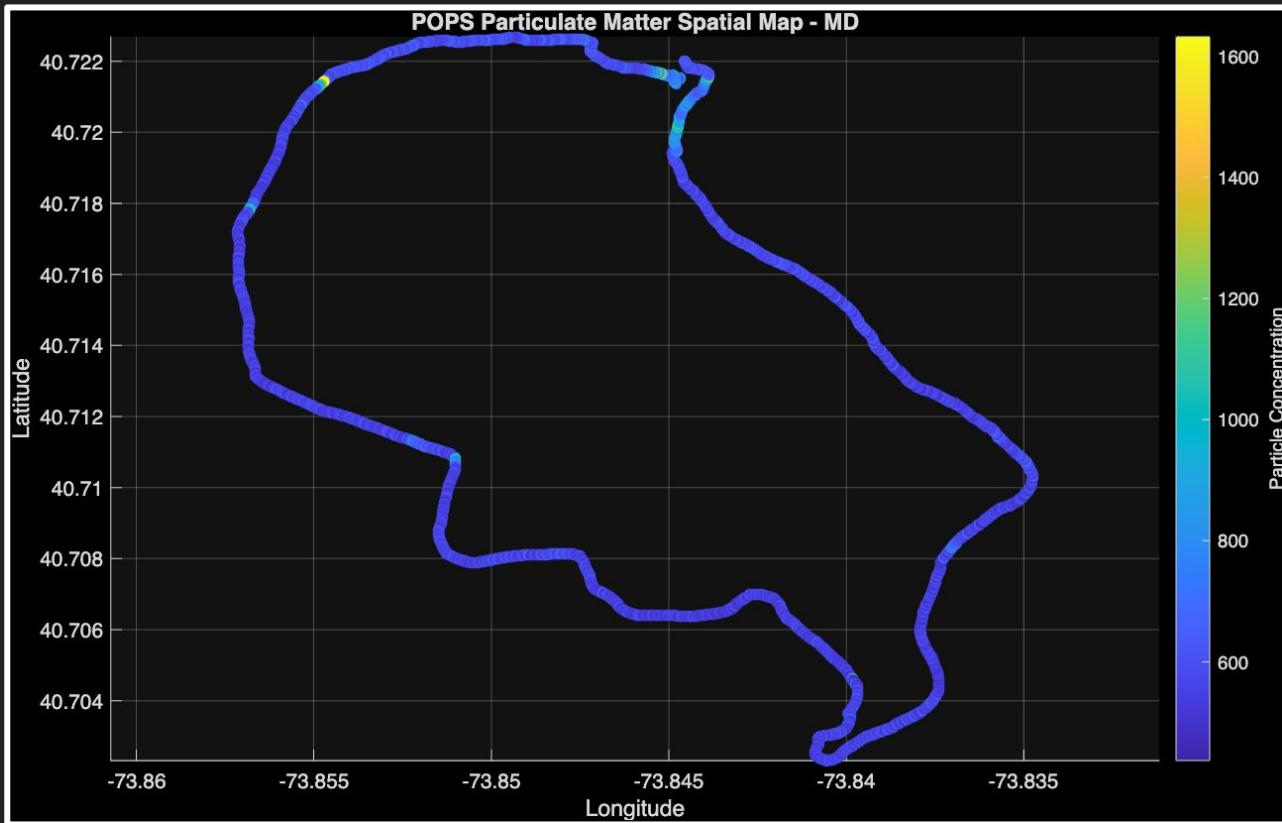
CO2 & Ozone



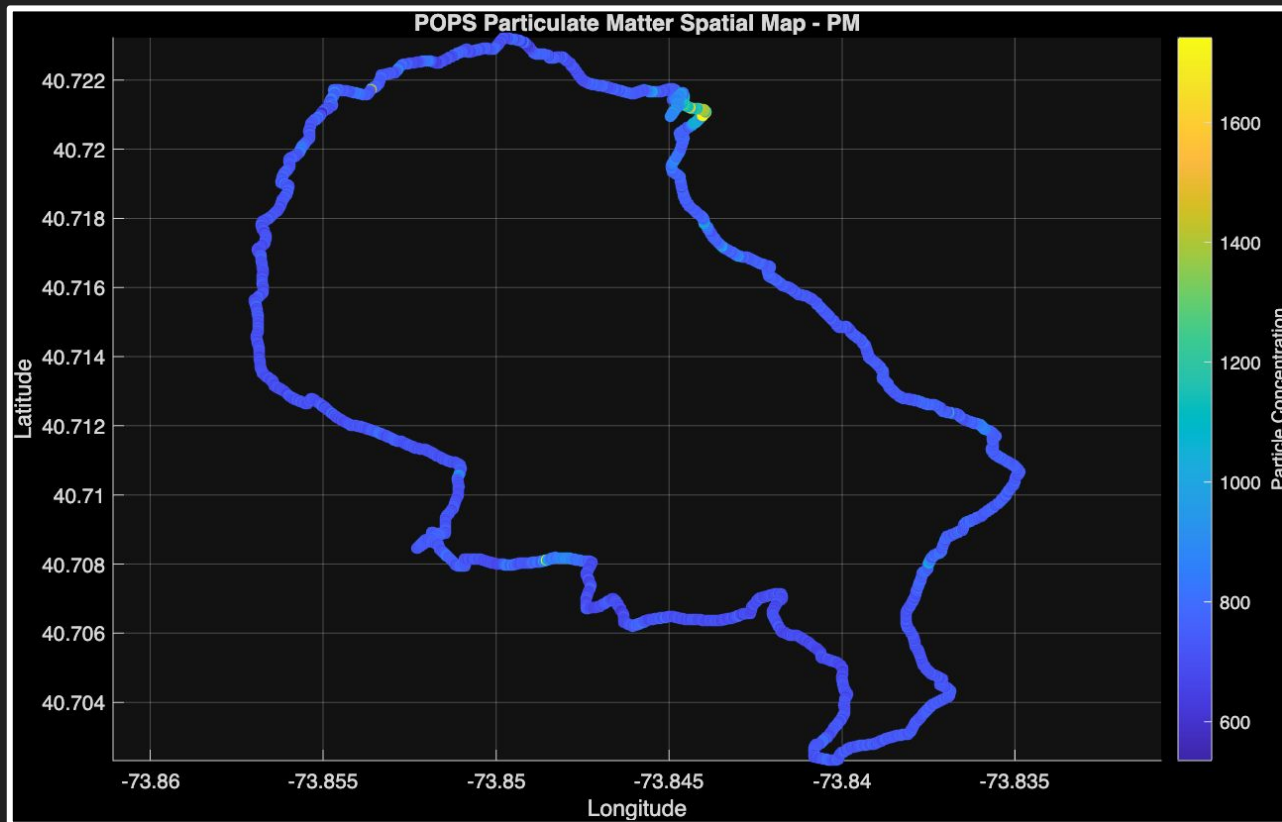
Particulate Matter



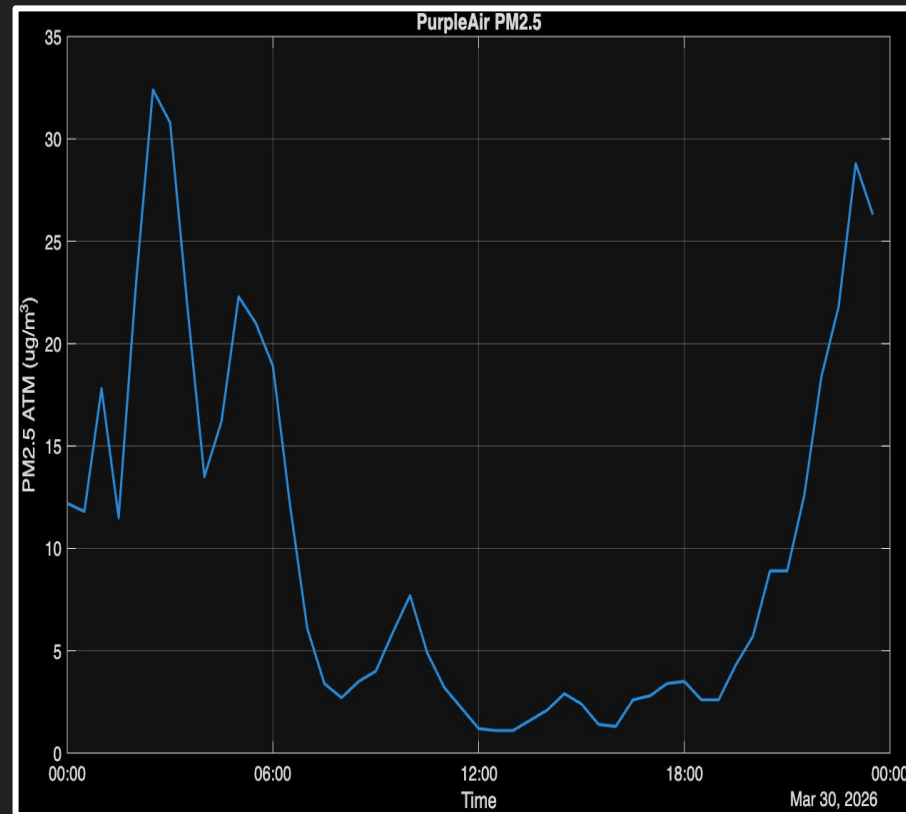
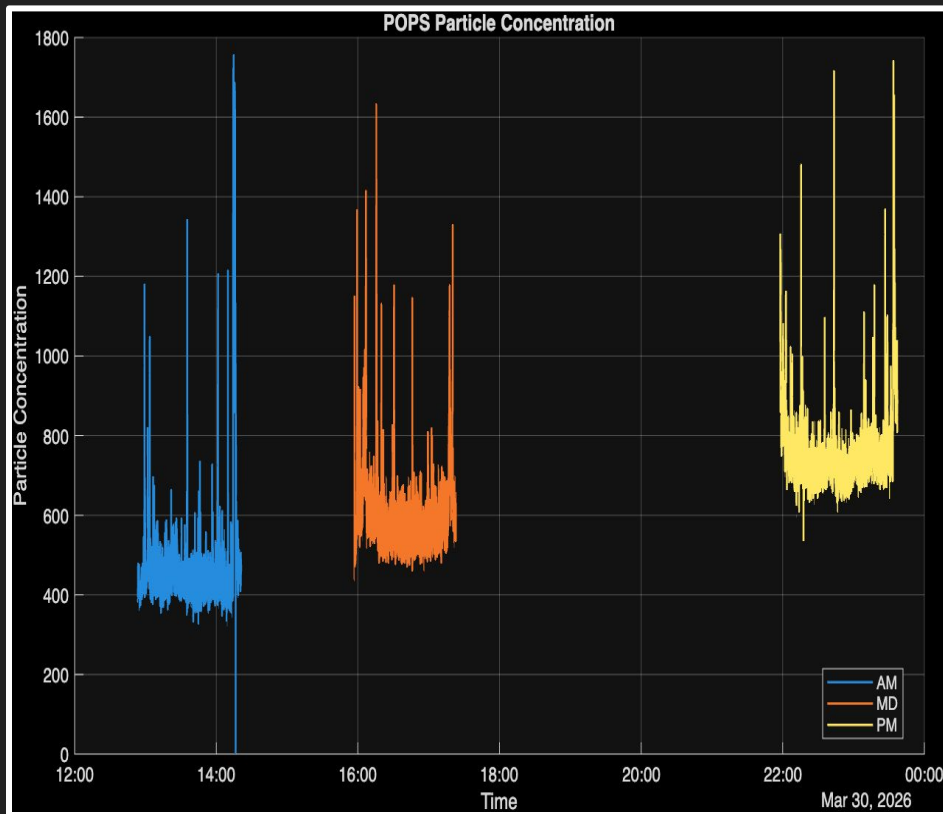
Particulate Matter



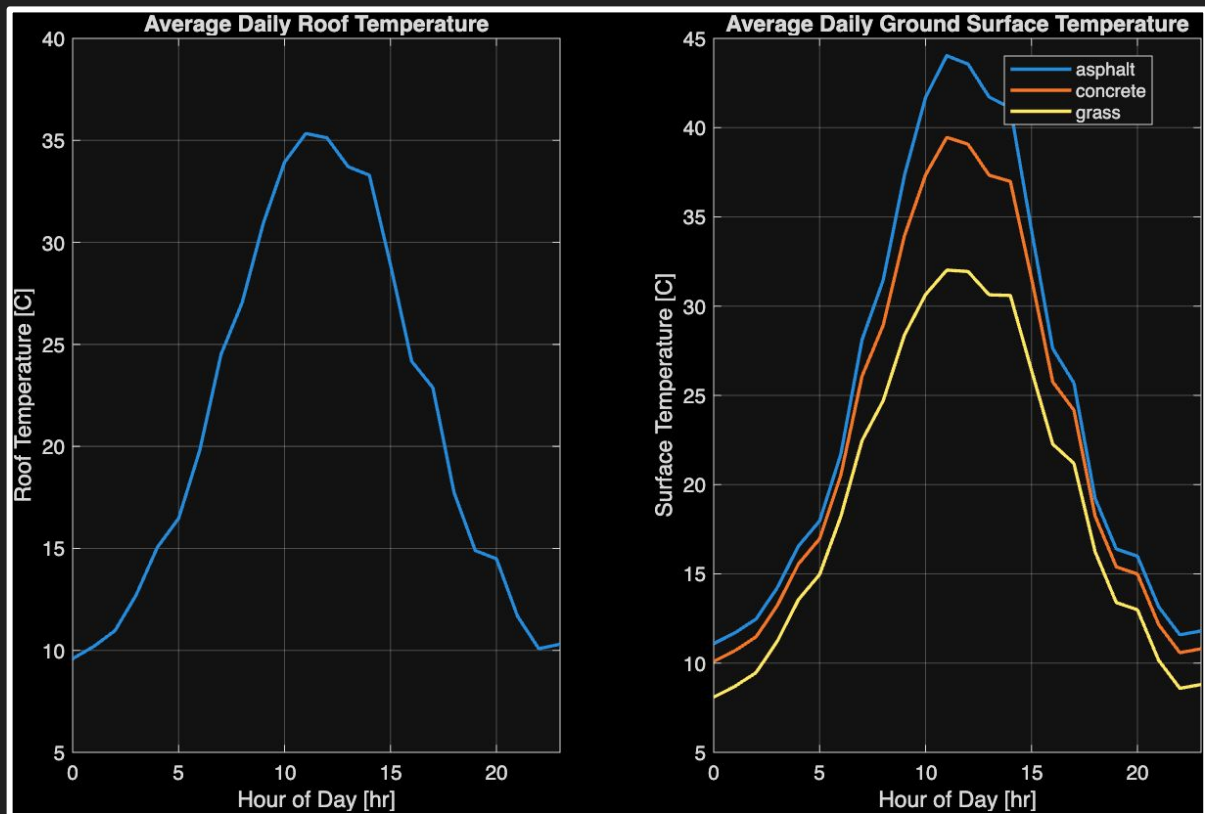
Particulate Matter



Particulate Matter

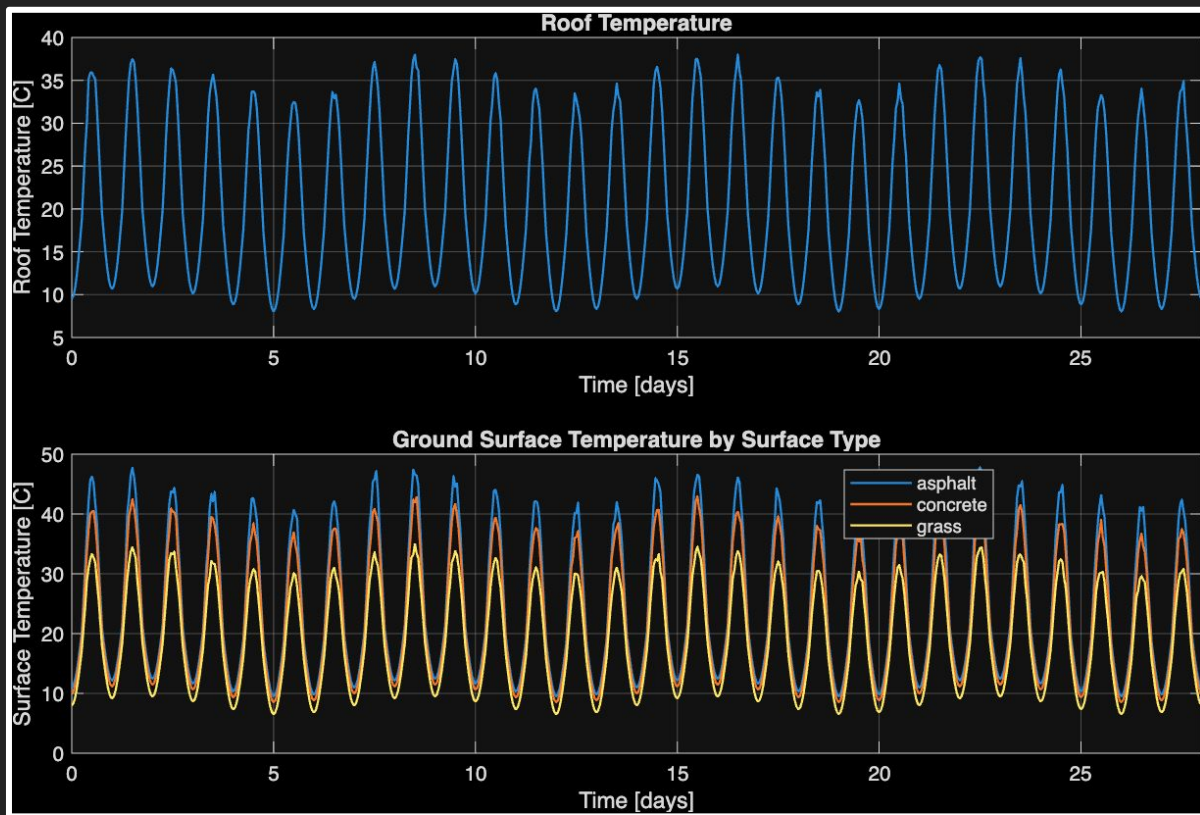


Average Daily Roof and Surface Temperature

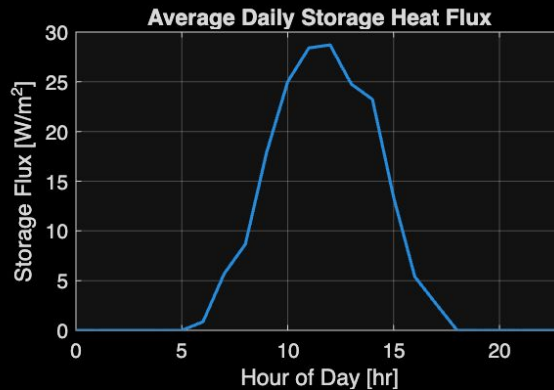
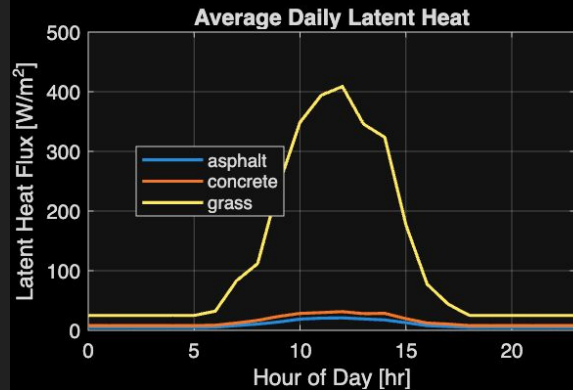
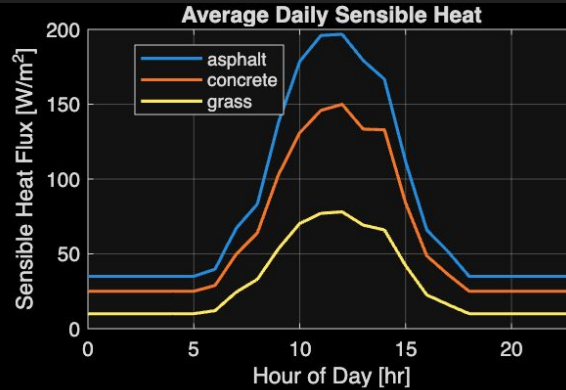
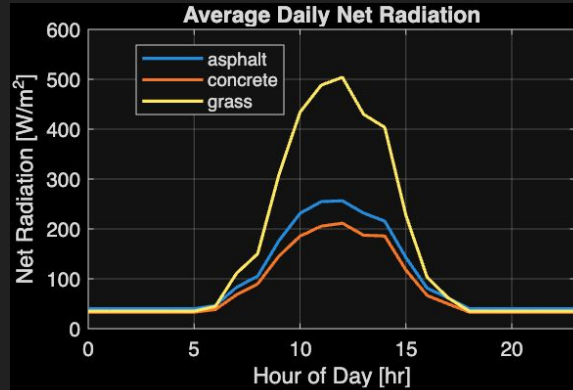


Roof temperature peaks around 35 °C whereas surface temperature, specifically asphalt, peaks around 44°C.

Monthly Roof and Surface Temperature



Average Daily Latent & Sensible Heat Flux



Sensible Heat peaked at 196 W/m^2 . Latent peaked at 406 W/m^2 . Daily Storage was at 29 W/m^2

Monthly Latent & Sensible Heat Flux

